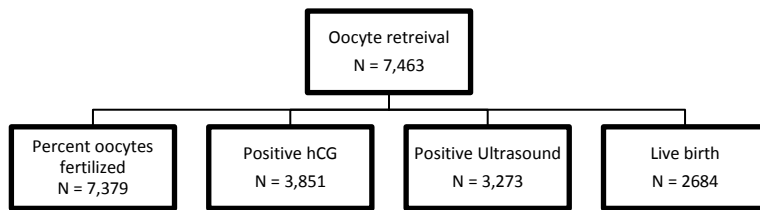
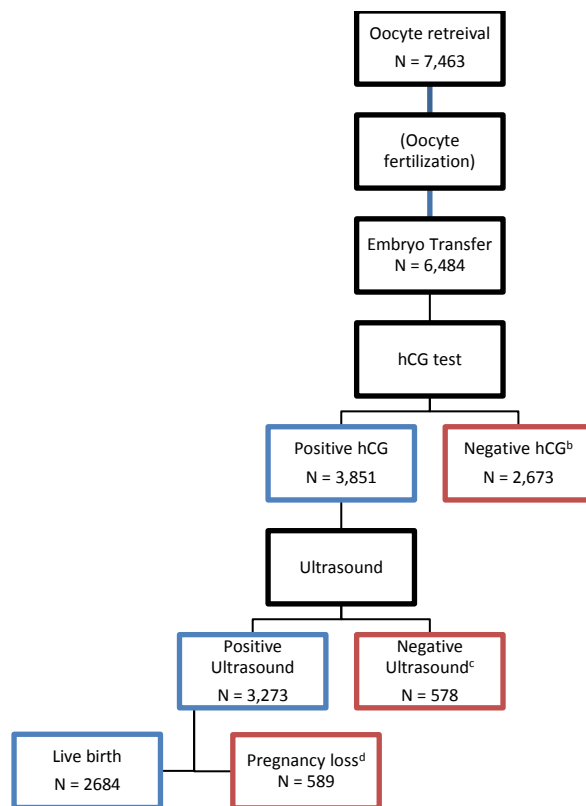


Supplemental Figure 1. Study Population and Outcomes

a. Likelihood of markers of successful IVF within overall study population^a:



b. Likelihood of poor IVF outcomes according to sequential risk sets:



^aLikelihood of increase in percent fertilized, positive hCG, positive ultrasound (visible gestational sac), and live birth among the entire study population.

^bLikelihood of negative hCG among those with an embryo transfer

^cLikelihood of negative ultrasound (no visible gestational sac) among those with a positive hCG

^dLikelihood of a pregnancy loss defined as no live birth following a positive ultrasound

Supplemental Table 1. Baseline and outcome characteristics by clinic

Participant Characteristic	All Participants			Seattle			San Francisco			Los Angeles			Shady Grove ^a		
	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD
Age at cycle start (years)	7,463	34.9	4.6	1,171	34.4	4.4	759	35.4	4.5	719	36.6	4.5	4,814	34.7	4.6
BMI at cycle start (kg/m ²)	7,463	25.3	5.1	1,171	25.1	5.0	759	25.1	4.9	719	23.3	4.5	4,814	25.6	5.2
Neighborhood SES index z-score	7,425	0.0	5.1	1,171	1.0	4.4	759	0.5	5.4	719	0.3	5.3	4,814	0.2	5.1
Percent oocytes fertilized (%)	7,379	67.5	23.6	1,162	64.5	23.8	748	77.6	20.1	707	72.3	24.4	4,762	65.9	23.4
	N		%	N		%	N		%	N		%	N		%
Ethnicity															
White	4,291		57.5	636		54.3	377		49.7	359		49.9	2,919		60.6
Non-white	3,172		42.5	535		45.7	382		50.3	360		50.1	1,895		39.4
Smoking history															
Yes	355		4.8	82		7.0	48		6.3	31		4.3	194		4.0
No	4,590		61.5	449		38.3	697		91.8	569		79.1	2,875		59.7
Unknown	2,518		33.7	640		54.7	14		1.8	119		16.6	1,745		36.3
Infertility diagnosis															
Male Infertility	2,219		29.7	432		36.9	273		36.0	131		18.2	1,383		28.7
Diminished Ovarian Reserve	1,305		17.5	236		20.2	257		33.9	138		19.2	674		14.0
Ovulation disorders/PCO	1,050		14.1	162		13.8	123		16.2	62		8.6	703		14.6
Tubal factors	961		12.9	182		15.5	113		14.9	49		6.8	617		12.8
Endometriosis	431		5.8	92		7.9	48		6.3	30		4.2	261		5.4
Uterine Factor	276		3.7	46		3.9	49		6.5	20		2.8	161		3.3
Unexplained	1,674		22.4	148		12.6	113		14.9	117		16.3	1,296		26.9
Other	923		12.4	116		9.9	90		11.9	203		28.2	514		10.7
Near roadway residence ^b	893		12.0	147		12.6	104		13.7	119		16.6	523		10.9
IVF outcomes															
Embryo grade, Good	5,397		72.3	836		71.4	408		53.8	431		59.9	3,722		77.3
Positive hCG	3,811		51.1	621		53.0	423		55.7	278		38.7	2,489		51.7
Positive ultrasound	3,273		43.9	497		42.4	355		46.8	261		36.3	2,160		44.9
Live birth	2,684		36.0	407		34.8	296		39.0	212		29.5	1,769		36.8

^aShady Grove includes: Rockville, MD; Baltimore, MD; Chesterbrook, PA^bDistance to roadway (DTR) dichotomized as near roadway residence versus not near road residence. Near roadway defined as within 100m of an A1 or A2, or within 50m of an A3 class roadway.

Supplemental Table 2a. Association of air pollution^a levels with markers of successful IVF, minimal model adjustment

	N	Model 1 ^a	
		Change in %	95% CI
<i>Percent oocytes fertilized</i>	7,379		
PM _{2.5}		0.18	(-0.013, 0.371)
PM ₁₀		0.09	(-0.006, 0.185)
NO ₂		0.04	(0.010, 0.071)
	N	RR	95% CI
<i>Embryo grade - Good</i>	7,463		
PM _{2.5}		0.99	(0.97, 1.01)
PM ₁₀		0.95	(0.93, 0.96)
NO ₂		0.95	(0.92, 0.96)
<i>Positive hCG test</i>	7,463		
PM _{2.5}		0.95	(0.91, 0.97)
PM ₁₀		0.96	(0.94, 0.98)
NO ₂		0.96	(0.92, 0.98)
<i>Positive ultrasound</i>	7,463		
PM _{2.5}		0.98	(0.94, 1.02)
PM ₁₀		0.98	(0.95, 1.01)
NO ₂		0.97	(0.93, 1.00)
<i>Live birth</i>	7,463		
PM _{2.5}		0.98	(0.93, 1.02)
PM ₁₀		0.98	(0.95, 1.01)
NO ₂		0.96	(0.91, 0.99)

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-μg/m³, PM₁₀ per 4-μg/m³, NO₂ per 5-ppb change)

Supplemental Table 2b. Association of air pollution^a levels with markers of poor IVF, minimal model adjustment

	N	RR	Model 1 ^a 95% CI
<i>Negative hCG</i>	6,484		
PM _{2.5}		1.03	(0.98, 1.06)
PM ₁₀		1.00	(0.97, 1.02)
NO ₂		1.00	(0.95, 1.04)
<i>Negative ultrasound</i>	3,851		
PM _{2.5}		0.81	(0.72, 0.89)
PM ₁₀		0.87	(0.79, 0.94)
NO ₂		0.94	(0.84, 1.04)
<i>Pregnancy loss</i>	3,273		
PM _{2.5}		1.04	(0.93, 1.14)
PM ₁₀		1.01	(0.94, 1.08)
NO ₂		1.06	(0.95, 1.16)

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-μg/m³, PM₁₀ per 4-μg/m³, NO₂ per 5-ppb change)

Supplemental Table 3. Association of air pollution with IVF outcomes stratified by clinic location^{a,b}

	Rockville			Los Angeles			San Francisco			Seattle			Baltimore/ Chesterbrook		
	N=3,949			N=719			N=759			N=1,171			N=865		
	RR	95% CI		RR	95% CI		RR	95% CI		RR	95% CI		RR	95% CI	
		Lower	Upper		Lower	Upper		Lower	Upper		Lower	Upper		Lower	Upper
<i>Embryo grade - Good</i>															
PM _{2.5}	0.99	0.95	1.03	1.04	0.93	1.16	0.91	0.82	1.01	1.02	0.96	1.08	0.95	0.87	1.05
PM ₁₀	1.01	0.98	1.04	1.01	0.94	1.09	0.95	0.88	1.02	1.04	1.00	1.08	0.94	0.88	1.01
NO ₂	0.99	0.97	1.02	1.05	0.96	1.15	0.91	0.81	1.02	0.98	0.91	1.05	0.98	0.92	1.04
<i>Positive hCG</i>															
PM _{2.5}	0.95	0.88	1.01	1.12	0.96	1.32	0.91	0.83	1.01	1.09	1.00	1.19	0.95	0.82	1.11
PM ₁₀	0.97	0.92	1.02	1.04	0.93	1.17	0.91	0.85	0.98	1.08	1.02	1.14	0.96	0.87	1.06
NO ₂	0.99	0.94	1.04	1.11	0.97	1.26	0.90	0.81	1.01	0.96	0.87	1.06	1.01	0.91	1.11
<i>Positive ultrasound</i>															
PM _{2.5}	0.94	0.87	1.02	1.19	1.00	1.42	0.91	0.80	1.03	1.09	0.98	1.21	0.88	0.74	1.05
PM ₁₀	0.98	0.92	1.04	1.08	0.95	1.21	0.92	0.85	1.00	1.07	1.00	1.15	0.93	0.82	1.05
NO ₂	0.98	0.93	1.05	1.18	1.03	1.36	0.88	0.77	1.00	0.92	0.81	1.04	0.98	0.87	1.10
<i>Live birth</i>															
PM _{2.5}	0.93	0.85	1.02	1.21	0.99	1.48	0.89	0.78	1.03	1.03	0.91	1.17	0.87	0.70	1.07
PM ₁₀	0.99	0.92	1.06	1.10	0.95	1.26	0.93	0.84	1.02	1.01	0.92	1.10	0.90	0.77	1.06
NO ₂	0.96	0.89	1.03	1.18	1.01	1.38	0.89	0.77	1.05	0.88	0.76	1.01	0.95	0.83	1.09

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-μg/m³, PM₁₀ per 4-μg/m³, NO₂ per 5-ppb change)^bModel adjustment: age, BMI, race, NSES

Supplemental table 4. Association of air pollution levels^a with live birth and other IVF outcomes^b
Excluding: Single female and Same sex partner participants^c

	N	Beta coefficient	95% CI, Lower limit	95% CI, Upper limit
<i>Percent oocytes fertilized</i>	7314			
PM _{2.5}		-0.28	-0.56	-0.01
PM ₁₀		-0.14	-0.28	0.00
NO ₂		-0.05	-0.09	-0.02
Near roadway		-0.21	-1.87	1.44
	N	RR	95% CI, Lower limit	95% CI, Upper limit
<i>Embryo quality - Good</i>	7314			
PM _{2.5}		0.99	0.96	1.02
PM ₁₀		1.01	0.98	1.03
NO ₂		0.99	0.97	1.02
Near roadway	7314	0.97	0.93	1.02
<i>Positive hCG test</i>				
PM _{2.5}		0.98	0.94	1.02
PM ₁₀		0.99	0.96	1.02
NO ₂		0.99	0.95	1.02
Near roadway		0.90	0.84	0.97
<i>Positive ultrasound</i>	7314			
PM _{2.5}		0.98	0.93	1.03
PM ₁₀		0.99	0.96	1.03
NO ₂		0.98	0.94	1.02
Near roadway		0.90	0.83	0.98
<i>Live birth</i>	7314			
PM _{2.5}		0.96	0.90	1.02
PM ₁₀		0.98	0.94	1.03
NO ₂		0.96	0.91	1.01
Near roadway		0.90	0.82	1.00
<i>Negative hCG</i>	6346			
PM _{2.5}		1.01	0.95	1.08
PM ₁₀		1.03	0.99	1.08
NO ₂		1.01	0.97	1.06
Near roadway		1.12	1.03	1.22
<i>Negative ultrasound</i>	3770			
PM _{2.5}		1.02	0.88	1.18
PM ₁₀		0.98	0.88	1.09
NO ₂		1.05	0.92	1.20
Near roadway		1.02	0.80	1.31
<i>Pregnancy loss</i>	3205			
PM _{2.5}		1.09	0.94	1.27
PM ₁₀		1.06	0.96	1.18
NO ₂		1.10	0.97	1.24
Near roadway		0.96	0.76	1.22

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-µg/m³, PM₁₀ per 4-µg/m³, NO₂ per 5-ppb change)

^bModel adjustments: age, BMI, race, NSES, clinic location

^cExcluding n=100 who were listed as "Single Female" and did not have other designations of infertility and n=49 who were listed as "Same Sex Partner" and did not have any other designations of infertility.

Supplemental table 5. Association of air pollution levels^a with live birth and other IVF outcomes^bExcluding: Male infertility^c

	N	Beta coefficient	95% CI, Lower limit	95% CI, Upper limit
<i>Percent oocytes fertilized</i>	6085			
PM _{2.5}		-0.176	-0.483	0.131
PM ₁₀		-0.088	-0.241	0.065
NO ₂		-0.028	-0.067	0.011
Near roadway		-0.485	-2.292	1.322
	N	RR	95% CI, Lower limit	95% CI, Upper limit
<i>Embryo quality - Good</i>	6085			
PM _{2.5}		0.98	0.95	1.01
PM ₁₀		1.00	0.97	1.02
NO ₂		0.99	0.97	1.02
Near roadway		0.98	0.93	1.03
<i>Positive hCG test</i>				
PM _{2.5}		0.98	0.93	1.03
PM ₁₀		0.98	0.95	1.02
NO ₂		0.99	0.95	1.03
Near roadway		0.90	0.83	0.98
<i>Positive ultrasound</i>				
PM _{2.5}		0.97	0.91	1.02
PM ₁₀		0.99	0.95	1.03
NO ₂		0.98	0.93	1.02
Near roadway		0.89	0.81	0.98
<i>Live birth</i>				
PM _{2.5}		0.94	0.88	1.01
PM ₁₀		0.98	0.94	1.03
NO ₂		0.95	0.90	1.01
Near roadway		0.88	0.78	0.98
<i>Negative hCG</i>	5212			
PM _{2.5}		1.02	0.95	1.09
PM ₁₀		1.03	0.99	1.08
NO ₂		1.01	0.95	1.06
Near roadway		1.12	1.02	1.22
<i>Negative ultrasound</i>	3050			
PM _{2.5}		1.06	0.89	1.25
PM ₁₀		0.96	0.85	1.09
NO ₂		1.10	0.94	1.28
Near roadway		1.11	0.86	1.45
<i>Pregnancy loss</i>	2609			
PM _{2.5}		1.14	0.97	1.35
PM ₁₀		1.06	0.94	1.19
NO ₂		1.12	0.98	1.27
Near roadway		1.03	0.81	1.32

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-μg/m³, PM₁₀ per 4-μg/m³, NO₂ per 5-ppb change)^bModel adjustments: age, BMI, race, NSES, clinic location^cExcluding n=1,378 who had a diagnosis specifying male infertility factor did not have other designations of infertility.

Supplemental Table 6. Association of air pollution levels with IVF outcomes comparing spatiotemporal and spatial national model exposure estimates^a

N=4,012	Spatiotemporal estimates			National estimates ^b			Difference in CI width
	RR	95% CI		RR	95% CI		
		Lower	Upper		Lower	Upper	
<i>Live birth</i>							
PM _{2.5}	0.95	0.87	1.04	0.97	0.86	1.10	0.07
NO ₂	0.98	0.92	1.04	0.98	0.91	1.05	0.02
<i>Embryo grade - Good</i>							
PM _{2.5}	0.97	0.93	1.01	1.03	0.97	1.09	0.04
NO ₂	1.02	0.99	1.04	1.02	0.99	1.05	0.01
<i>Positive hCG</i>							
PM _{2.5}	0.94	0.88	1.00	0.97	0.88	1.07	0.06
NO ₂	0.99	0.94	1.03	0.99	0.94	1.05	0.02
<i>Positive ultrasound</i>							
PM _{2.5}	0.96	0.89	1.04	0.99	0.89	1.10	0.07
NO ₂	1.00	0.95	1.05	1.00	0.94	1.07	0.02

^aAir pollution levels reported per unit IQR (PM_{2.5} per 2-μg/m³, PM₁₀ per 4-μg/m³, NO₂ per 5-ppb change). Model adjustments: age, BMI, race, clinic location, NSES.

^bSpatiotemporal estimates available for n=4,012 participants, comparison national spatial model estimates presented using the same population subset.